



ASSIGNMENT No: 13

TITLE: Write a Java program to perform reading from and writing to a file using file handling operations.

Problem Statement

Write a Java program to perform reading from and writing to a file using file handling operations.

Learning Objectives

To perform file reading and writing operations using Java file handling classes.

Theory

Java provides file handling through the java.io package. Classes such as File, FileReader, FileWriter, BufferedReader, and BufferedWriter are used to perform input and output operations. File handling enables permanent storage and retrieval of application data. Proper exception handling is necessary to manage file-related errors. These concepts are essential for data persistence.

```
import java.io.*;

public class Main {
    Run | Debug
    public static void main(String[] args) {
        try {
            FileWriter fw = new FileWriter(fileName: "Exp13/data.txt");
            fw.write(str: "Hello, this is a Java file handling program.");
            fw.close();

            FileReader fr = new FileReader(fileName: "Exp13/data.txt");
            BufferedReader br = new BufferedReader(fr);

            String line;

            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();
            fr.close();

        } catch (IOException e) {
            System.out.println("File error: " + e.getMessage());
        }
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Java-Questions> java -cp Exp13 Main
Hello, this is a Java file handling program.
PS C:\Users\DHARMIT SHAH\Java-Questions> █
```

Exercise

- 1 Create a program to write and read employee details from a file.
- . Create a Java program to write patient details (patient ID, name, age, and
- 2 diagnosis) to a file and read the details from the file for display.
- .

```
import java.io.*;

public class Exp1 {
    public static void main (String[] args) {
        try {
            FileWriter fw = new FileWriter("Exp13/Exercise/employee.txt");

            fw.write("Employee ID: 1010\n");
            fw.write("Name: Raman\n");
            fw.write("Salary: 60000\n");

            fw.close();

            FileReader fr = new FileReader("Exp13/Exercise/employee.txt");
            BufferedReader br = new BufferedReader(fr);

            String line;

            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();
            fr.close();

        } catch (IOException e) {
            System.out.println("File error: " + e.getMessage());
        }
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Java-Questions> javac Exp13/Exercise/Exp1.java
>> java Exp13.Exercise.Exp1
>>
Employee ID: 1010
Name: Raman
Salary: 60000
PS C:\Users\DHARMIT SHAH\Java-Questions>
```

```
package Exp13.Exercise;

import java.io.*;

public class Exp2 {
    public static void main (String[] args) {
        try {
            FileWriter fw = new FileWriter("Exp13/Exercise/patient.txt");
            fw.write("Patient ID: 101\n");
            fw.write("Name: Amit\n");
            fw.write("Age: 25\n");
            fw.write("Diagnosis: Fever\n");

            fw.close();

            FileReader fr = new FileReader("Exp13/Exercise/patient.txt");
            BufferedReader br = new BufferedReader(fr);

            String line;

            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();
            fr.close();

        } catch (IOException e) {
            System.out.println("File error: " + e.getMessage());
        }
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Java-Questions> javac Exp13/Exercise/Exp2.java
>> java Exp13.Exercise.Exp2
Patient ID: 101
Name: Amit
Age: 25
Diagnosis: Fever
PS C:\Users\DHARMIT SHAH\Java-Questions> █
```

Conclusion

This experiment helped in understanding the concept of file handling in Java using the java.io package. It demonstrated how to perform reading and writing operations using classes such as FileReader, FileWriter, BufferedReader, and BufferedWriter. The experiment also showed how file handling can be used for permanent storage and retrieval of data. Overall, it improved understanding of data persistence and the importance of handling file-related errors properly.